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Melt free-radical grafting of glycidyl methacrylate onto polypropylene

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SUMMARY:

This paper concerns free-radical grafting of glycidyl methacrylate (GMA) onto polypropylene (PP) in the molten state carried out in a batch mixer. This free-radical grafting proceeds very rapidly as it goes to completion within two to three minutes at 200 °C. The grafting yield of GMA alone is low and it is greatly enhanced upon addition of styrene as a second monomer with a reduced chain degradation of PP. Homopolymer and copolymer of GMA formed during grafting were quantified. It is suggested that in the presence of styrene, the dominating grafting mechanism is that styrene reacts with the PP macroradicals to form more stable styryl macroradicals which then react with GMA to form GMA-grafted PP. The rate of this reaction is much greater than that between the PP macroradicals and GMA. Unlike grafting of mixtures of maleic anhydride and styrene, which is dictated primarily by formation of a charge transfer complex, free-radical grafting of GMA in the presence of styrene follows a random copolymerization process.

ZUSAMMENFASSUNG:

Geschmolzenes Polypropylen (PP) wurde in einem Batchmischer radikalisch mit Glycidylmethacrylat (GMA) gepfropft. Bei 200 °C verläuft die Pfropfung sehr rasch und ist nach 2 bis 3 Minuten abgeschlossen. Die Pfropfausbeute beim ausschließlichen Einsatz von GMA ist niedrig und kann in Gegenwart von Styrol als Comonomerem erhöht werden, wobei auch der Kettenabbau des PP verringert wird. Die während der Pfropfung gebildeten Mengen an homo- und copolymerem GMA wurden bestimmt. Es kann angenommen werden, daß in Gegenwart von Styrol dieses zunächst mit den PP-Makroradikalen reagiert und stabilere Styryl-Radikale bildet, die dann mit GMA gepfropft werden. Die Geschwindigkeit dieser Reaktion ist sehr viel größer als die der Reaktion von GMA mit PP-Makroradikalen. Im Gegensatz zu der über einen Charge-Transfer-Komplex ablaufenden Pfropfung von Mischungen aus Maleinsäureanhydrid und Styrol folgt die Pfropfung von GMA in Gegenwart von Styrol einem statistischen Copolymerisationsprozeß.

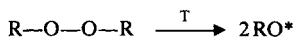
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Introduction

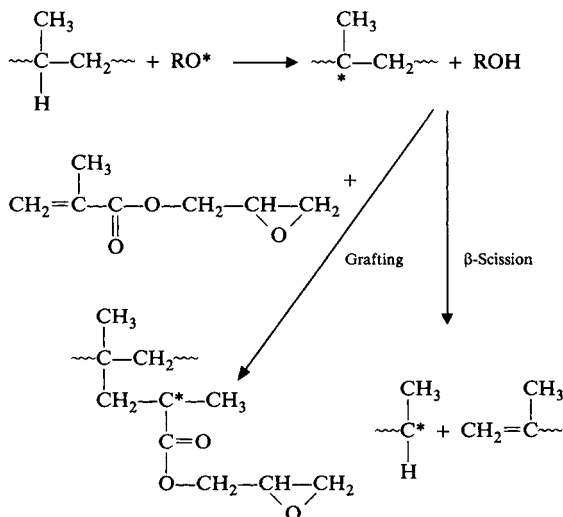
Free-radical grafting of monomers onto polyolefins has gained wide industrial application. One of them is to functionalize polyolefins for subsequent reactive blending with other polymers in order to form compatibilized polymer blends with improved properties and morphological stability¹⁻⁴. Among the most frequently employed monomers are unsaturated carboxylic derivatives, such as maleic and itaconic anhydrides, and vinylic and acrylic substances containing a second functionality^{5,6}. As the reactivity of polyolefins stems from their hydrogen atoms along the hydrocarbon skeleton that are subjected to free radical attack, aside from desired grafting, crosslinking or degradation may occur. Polyethylene and ethylene-propylene copolymers are prone to crosslinking⁶, whereas polypropylene (PP) suffers from chain degradation by β -scission^{7,8}. Additional difficulty of free radical grafting arises from the competition between monomer grafting and homopolymerization. Limited solubility of monomers in polyolefin melts is also an important factor that controls the overall grafting⁷.

During the free radical grafting of GMA onto PP the following reactions can occur:

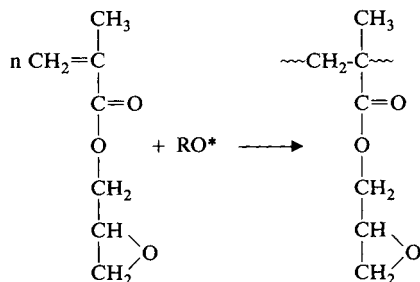
1. Free radical formation



2. Hydrogen abstraction, grafting and degradation



3. Homopolymerization



If another monomer is present, e. g. styrene, it may also react with PP and/or copolymerize with GMA.

The macroradicals once formed on the PP backbone will undergo either chain degradation or grafting. Due to the high unstability of tertiary macroradicals, the degradation process is extremely fast at elevated temperatures. Thus, effectively stabilizing and/or consuming macroradicals in favor of grafting are almost the only means to reduce chain degradation and homopolymerization under melt processing conditions.

The purpose of this work was devoted to the grafting of glycidyl methacrylate monomer (GMA) onto polypropylene (PP) through free-radical chemistry with high grafting yields and reduced chain degradation. This was achieved by adding a second monomer, styrene (St), to the grafting system. The mechanism of the free-radical grafting of GMA onto PP in the presence or absence of styrene was investigated. The ultimate goal was to use the reactivity of the GMA moieties grafted onto PP for in situ compatibilization of PP-based polymer blends. This will be reported in forthcoming papers.

Experimental

Materials

The PP used in this study was a commercial grade product (Valtec CL101D) of Himont (Belgium) in the form of porous pellets. This porous PP was capable of absorbing much larger amounts of liquid monomers than the one in the form of normal pellets, thus facilitating material feeding and reducing monomer loss. The number and weight-average molecular weights of the PP were measured by GPC: $\bar{M}_n$ 141 000, $\bar{M}_w$ 645 000, MFI 0.3 g/10 min (ASTM 1238L, 230 °C and 2.16 kg). GMA and styrene were purchased from Aldrich. 2,5-Dimethyl-2,5-di(tert-butylperoxy)hexane (DHBP)

and 1,3-bis(tert-butylperoxy-isopropyl)benzene (Perk-14), commercial products of Akzo in a liquid state with a purity of 90%, were used as the peroxides. The half-lifetime of these two peroxides is close at 200 °C, being about 30 s.

Grafting procedures

The grafting was carried out in a Rheocord Haake mixer (50 cm³) in which two sigma rotors rotated in opposite directions at 64 r. p. m. For a typical experimental run, the PP pellets (45 g) along with liquid monomer(s) and peroxide were mechanically mixed in a cup for about 15 min in order that the liquid reagents were absorbed by the PP pellets. They were then charged to the mixer which was preheated up to 200 °C. Samples were taken from the mixing chamber after 15 min of reaction and quenched in liquid nitrogen in order to stop further reactions. Under certain circumstances, samples were also taken after 2 or 3 min of reaction.

Grafting yield and molecular weights

Modified PP samples were purified in two ways: they were either dissolved in hot xylene and then precipitated in methanol at room temperature; or they were dissolved in hot xylene and then precipitated in acetone at room temperature. The use of methanol allowed to precipitate not only the modified PP but also poly(glycidyl methacrylate), polystyrene and poly(glycidyl methacrylate-co-styrene) which might be formed during grafting. On the other hand, PP precipitated well in acetone which was capable of dissolving certain amounts of poly(glycidyl methacrylate), polystyrene and

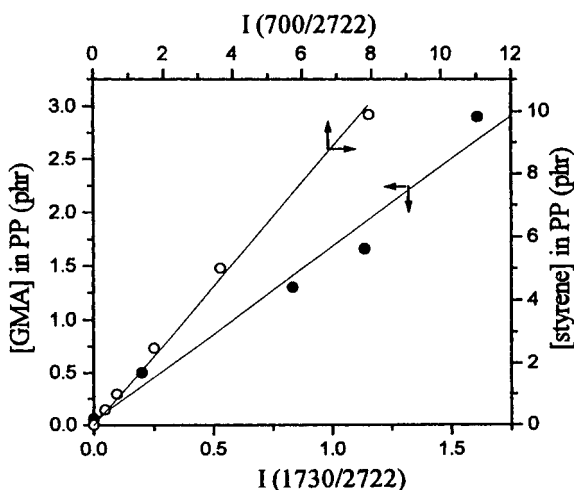


Fig. 1. IR calibration curves for determining the grafting yields of GMA (●) and styrene (○).

poly(glycidyl methacrylate-co-styrene). Those two ways of purifying modified PP samples allowed to determine, with some uncertainties though, the relative importance of homo- and/or copolymerization under the said grafting conditions.

The extent to which the molecular weight of the PP was reduced was characterized by GPC at high temperatures. Grafting yields were determined by FTIR using thin films (between 100 and 120 μm) of the purified PP samples. The peak of the PP at 2722 cm^{-1} was chosen as the reference, the one at 1730 cm^{-1} corresponding to grafted GMA and that at 700 cm^{-1} corresponding to grafted styrene were chosen as the targets. Calibrations were made for GMA and styrene, respectively, using thin films of known GMA and styrene contents (Fig. 1).

Results

Free radical reactivity of GMA, styrene, and their mixtures towards PP

When GMA alone reacted with PP in the presence of DHBP, the grafting yield was low and the molecular weights of PP decreased significantly. For example, a mixture of 5.8 phr (parts per hundred resin of PP) of GMA, 0.24 phr of DHBP and 100 phr of PP was mixed at about 200°C , the grafting yield of GMA was 0.38 phr, i. e., only 7% of the total GMA was grafted. When the concentration of DHBP was raised to 0.44 phr, 9% of the total GMA was grafted. Grafting yields of the same order of magnitude were also reported by Liu et al.⁴ under similar reaction conditions. Thus, it is not surprising that Gallucci and Going⁹ claimed that GMA could not be grafted onto PP using a melt grafting process.

The free-radical grafting reactivity of styrene alone towards PP appeared to be much greater than that of GMA alone. Accordingly, degradation of PP was

Tab. 1. Comparison of free radical grafting reactivity between GMA, styrene and their mixture^a.

$[\text{DHBP}]_i$ (phr)	$[\text{GMA}]_i$ (mhr)	$[\text{St}]_i$ (mhr)	$[\text{GMA}]_g$ (mhr)	$[\text{St}]_g$ (mhr)	$\bar{M}_w$	$\bar{M}_n$
0	0	0	0	0	314 000	105 000
0.2	0	0.043	0	0.014	283 000	91 000
0.2	0.043	0	0.003	0	229 000	80 000
0.2	0.043	0.043	0.010	0.006	327 000	99 000

^a $[\text{x}]_i$: initial moles of x per 100 g of PP resin, $[\text{x}]_g$: moles of x grafted onto 100 g of PP resin, mhr: moles per 100 g of PP resin.

more severe in the case of GMA. The reactivity of this latter increased greatly upon addition of styrene. This is illustrated in Tab. 1. When GMA alone (0.043 mol per hundred grams of PP (0.043 mhr)) reacted with PP for 15 min, the grafting yield was 0.003 mhr; when the same amount of styrene alone reacted with PP, its grafting yield was higher (0.014 mhr). Interestingly, when a mixture of 0.043 mhr of GMA and the same amount of styrene reacted with PP in the mixer for 15 min, the grafting yield of GMA was increased from 0.003 to 0.010 mhr whereas that of styrene was decreased from 0.014 to 0.006 mhr.

The competition between grafting and homopolymerization of styrene is represented in Fig. 2. After 15 min of reaction most of the reacted styrene was grafted onto PP. This was particularly so when the initial concentration of styrene $[St]_i$ was low: more than 70% of the reacted styrene was grafted onto PP when the initial concentration of styrene was below 4.4 phr; when $[St]_i$ was as high as 8 phr, homopolymerization became relatively important.

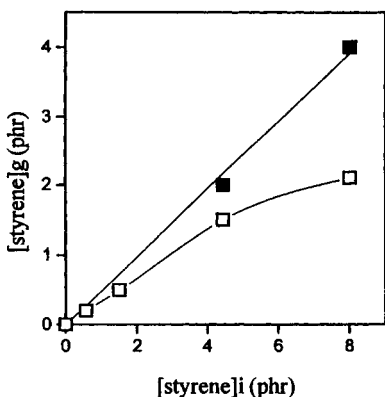


Fig. 2. Competition between grafting and homopolymerization of styrene ($[DHBP]_i = 0.2$ phr); (■) concentrations of polystyrene and grafted styrene in PP samples, (□) concentrations of grafted styrene alone.

Relationship between grafting yield of GMA and initial concentration of styrene

The grafting yields of GMA and styrene as a function of the initial molar ratio of $[St]_i$ over $[GMA]_i$ at particular concentrations of GMA (6.0 phr) and DHBP (0.20 phr) are shown in Fig. 3. Again, the grafting yield was low in the absence of styrene (0.38 phr) and it increased to 1.8 phr when $[St]_i/[GMA]_i$ was 1.5 mol/mol. When compared to the case without styrene addition, the number and weight-average molecular weights, $\bar{M}_w$ and $\bar{M}_n$, of the modified PP samples also increased, but to a lower extent than the grafting yield

Fig. 3. Grafting yields of GMA (●) and styrene (○) as a function of the initial molar ratio of $[St]_i/[GMA]_i$.

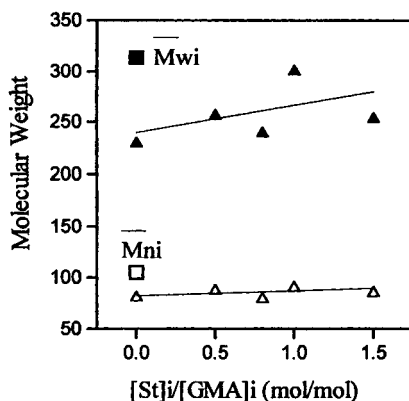
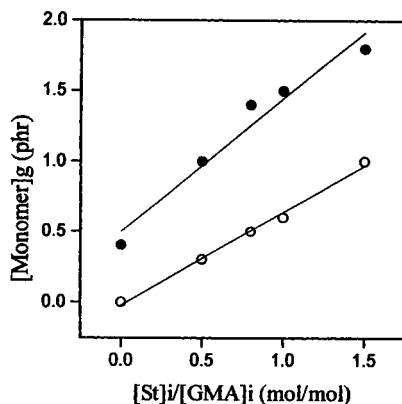


Fig. 4. Number (△) and weight-average (▲) molecular weights of the reacted PP as a function of the initial molar ratio of $[St]_i/[GMA]_i$; $\bar{M}_w$ (■) and $\bar{M}_n$ (□) denote the number and weight-average molecular weights of the PP alone after 15 min of mixing (note that chain degradation induced by thermo-mechanical mixing is important as the number and weight-average molecular weights of the virgin PP are 141 000 and 654 000 respectively).

(Fig. 4). According to the number-average molecular weights of the modified PP, 1.8 phr of GMA grafted onto PP means that on the average each PP chain contains about eleven GMA moieties.

Kinetics of the grafting of GMA, styrene and their mixture

The kinetics of the free radical grafting of a mixture of GMA and styrene in the batch mixer at about 200 °C is shown in Fig. 5. The grafting reaction proceeded very rapidly for both GMA and styrene monomers: the grafting

yields obtained after 2 and 3 min of reaction were already very close to those obtained after 15 min of reaction. In other words, the free radical grafting already went to completion within 3 min. Similar results were also observed when GMA or styrene alone reacted with PP. It should be pointed out that 2 or 3 min were virtually the minimum time required for generating meaningful data in the batch mixer.

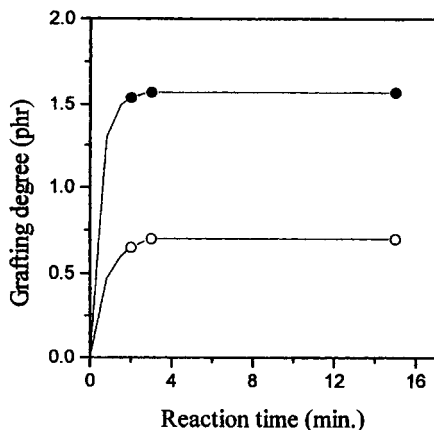


Fig. 5. Grafting yields of GMA (●) and styrene (○) as a function of mixing time in the batch mixer; $[DHB P]_i = 0.20$ phr, $[GMA]_i = 6$ phr, and $[St]_i/[GMA]_i = 1.0$ mol/mol.

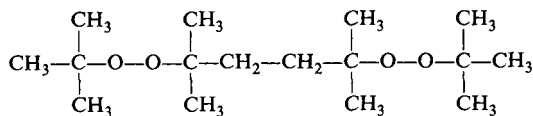
Effect of the nature and amount of peroxide on the grafting yield of GMA/styrene

In addition to DHB P, Perk-14 was used as a peroxide. This latter has a chemical structure similar to DHB P (Scheme 1) with a half-lifetime of about 30 s at 200 °C. Both peroxides are liquid at room temperature and soluble in a mixture of GMA and styrene. Thus, it was expected that the reactivity for these two peroxides would be close for free radical grafting of monomers.

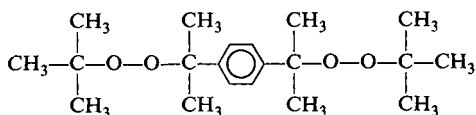
In order to compare the reactivity of these two peroxides, a mixture of GMA and styrene was grafted onto PP in the batch mixer at 200 °C for 15 min. The initial concentrations of the monomers were $[GMA]_i = 6.0$ phr and $[St]_i = 6.6$ phr. The concentration of each of the two peroxides varied from 0 to 0.44 phr. The grafting yield of GMA and that of styrene as a function of the initial peroxide concentration are shown in Fig. 6. Over the entire range of the peroxide concentration, the grafting yields of GMA and styrene were slightly

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Scheme 1. Chemical structures of DHBP and Perk-14.



DHBP



Perk-14

higher in the case of Perk-14. Also noteworthy is that the grafting yield of GMA or styrene increased with increasing peroxide concentration for both peroxides.

The number and weight-average molecular weights of the modified PP as a function of the initial concentration of Perk-14 are shown in Fig. 7. Overall,

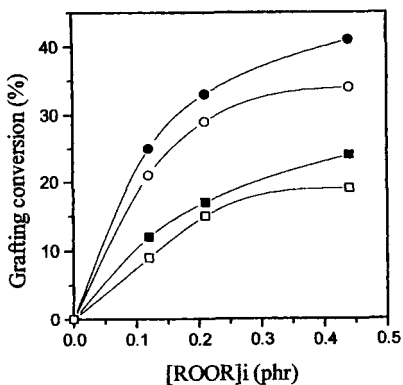


Fig. 6

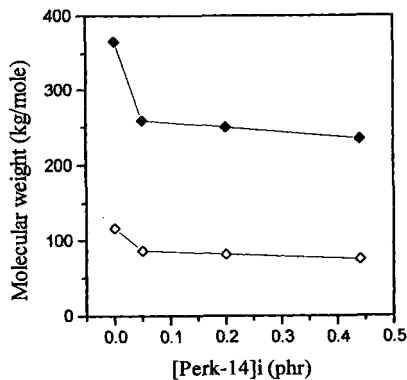


Fig. 7

Fig. 6. Comparison of grafting yields between DHBP and Perk-14; (●) grafting yield of GMA with Perk-14, (○) grafting yield of GMA with DHBP, (■) grafting yield of styrene with Perk-14, (□) grafting yield of styrene with DHBP; $[\text{St}]_i/[\text{GMA}]_i = 1.5 \text{ mol/mol}$.

Fig. 7. Number (◇) and weight-average (◆) molecular weights of the modified PP as a function of the concentration of Perk-14; $[\text{GMA}]_i = 6.0 \text{ phr}$, $[\text{St}]_i/[\text{GMA}]_i = 1.50 \text{ mol/mol}$.

$\bar{M}_w$ and $\bar{M}_n$ of the PP decreased with increasing peroxide concentration. Interestingly, the degree of chain degradation was already important with a small amount of peroxide (0.05 phr) and it was slightly more important with a further increase in peroxide concentration.

Discussion

The free radical grafting of GMA, styrene or mixtures of these two monomers is a rapid reaction, as grafting yields already reach a plateau within 2 or 3 min of mixing time (Fig. 5). The very high grafting rate is due primarily to a fast decomposition rate of the peroxide into primary free radicals (the half-lifetime of DHBP is about 30 s at 200°C). The short lifetime of the peroxide is also responsible for the existence of a plateau for the grafting yield. This is because the concentration of the primary free radicals is so high at the beginning of the grafting that there is a lack of the primary free radicals at longer times. As a result, the grafting process ceases.

On the other hand, the grafting yield of GMA or styrene increases with increasing peroxide concentration (Fig. 6) and the grafting yield does not level off with increasing peroxide concentration. This indicates that the final grafting yield of GMA is dictated by the effective concentration of the primary free radicals. Note that the ratio of the concentration of the peroxide to that of monomers, $[GMA + St]_i/[ROOR]_i$, for the grafting is between 40 and 150, which is almost of the same order of magnitude as for a typical polymerization. On the other hand, the conversion of GMA or styrene into the grafted one is by far less than 100%, between 10 and 40%. As the efficiency of a classical initiator is about 0.5 to 0.7 which is enough to convert monomers completely into polymers by a typical polymerization in solution, it is estimated that the initiator efficiency for the free radical grafting of GMA onto PP in the molten state is three to ten times lower, i. e. between 0.1 and 0.3. A much lower initiator efficiency in the molten PP medium is due to the fact that the viscosity of the medium is high. As a result, the cage effect is very important.

It is also noted that a small amount of peroxide is enough to degrade PP to an important extent, and a further increase in peroxide concentration causes only a slightly further degradation (Fig. 7). This implies that indeed the reaction between the primary free radicals and the tertiary hydrogen atoms of PP is very fast. Moreover, the resulting PP macroradicals are very unstable and decompose rapidly into two fragments by β -scission.

Additional interesting results are that the free radical reactivity of GMA alone towards PP is lower than that of styrene alone, but when the two monomers are mixed, the reactivity of GMA increases greatly whereas that of styrene decreases to the point that GMA becomes more reactive than styrene in the mixture (Tab. 1). This suggests that when styrene is present, the dominating grafting mechanism is that styrene reacts with the PP macroradicals to form more stable styryl macroradicals. These latter then react with GMA to form GMA-grafted PP. The rate of this reaction is much greater than that between the PP macroradicals and GMA. Unfortunately, at the present time, detailed mechanisms are not known due to the complications of the grafting reaction system. What can be said, however, is that the great contribution of styrene to the grafting of GMA must be related to the particular nature of styrene: its double bond of styrene is electronically conjugated with its benzene ring. As a result, its free radical reactivity and that in the corresponding free radical form must be different from a non-conjugated monomer.

Our previous work¹⁰ showed that when maleic anhydride (MA) was grafted onto PP, the presence of styrene also enhanced the grafting yield of MA greatly with less chain degradation of PP. This was proven to be due to the formation of a charge transfer complex (CTC) between the two monomers, which highly activated the otherwise weakly reactive double bond of MA towards free radicals. This argument apparently does not apply to the present system due to the following two experimental findings:

If there were formation of a charge transfer complex between GMA and styrene, the molar ratio of the grafted styrene to the grafted GMA would be expected to be equal to that of $[St]_i/[GMA]_i$. However, it was found that $[St]_g/[GMA]_g$ increased from 0.41 to 0.76 mol/mol when $[St]_i/[GMA]_i$ increased from 0.5 to 1.5 mol/mol (Fig. 3).

Secondly and more importantly, unlike mixtures of MA and styrene¹⁰, mixtures of GMA and styrene with various weight proportions were analyzed by nuclear magnetic resonance (NMR) and electron paramagnetic resonance (EPR) spectroscopy in the temperature range between room temperature and 120°C. No GMA/styrene complex was detected.

It turns out that the free radical grafting of a mixture of GMA and styrene onto PP is a random process. In a way, this agrees with the fact that the copolymerization between GMA and styrene is also a random process (the reactivity ratios of GMA and styrene are 0.53 and 0.44, respectively, at 60°C¹¹; and they are 0.63 and 0.34, respectively, at 65°C¹²). The enhanced grafting yield of GMA along with reduced chain degradation of PP in the

presence of styrene is due primarily to the conjugated double bond of styrene which reacts the PP macroradicals rapidly. The resulting styryl free radical is more stable with a longer half-lifetime allowing for subsequent reaction with GMA. In the absence of styrene, the PP macroradicals once formed, undergo chain degradation rapidly by β -scission so that fewer PP macroradicals will be available for subsequent monomer grafting.

Concluding remarks

The free radical reactivity of glycidyl methacrylate (GMA) towards molten polypropylene (PP) is weak. It can be greatly enhanced upon addition of styrene as a second monomer with a reduced chain degradation of PP due primarily to the particular character of the double bond of styrene, which is electronically conjugated with a benzene ring. It is suggested that when styrene is present, the dominating grafting mechanism is that styrene reacts with the PP macroradicals to form more stable styryl macroradicals. These latter then react with GMA to form GMA-grafted PP. The rate of this reaction is much greater than that between the PP macroradicals and GMA. Unlike grafting of mixtures of maleic anhydride and styrene onto PP, which is dictated by formation of a charge transfer complex, grafting of mixtures of GMA and styrene follows a random copolymerization process. The free radical grafting of GMA, styrene or their mixtures is a fast reaction as it ceases within 2 to 3 min at 200°C. The half-lifetime and the efficiency of peroxide is a critical parameter that governs the free radical grafting rate and yield. Further studies are still needed to gain more insights into the grafting mechanism.

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¹ A. J. Moffett, M. E. J. Dekkers, *Polym. Eng. Sci.* **32** (1992) 1

² EP 88300040.8 (1988), General Electric, Inv.: S. V. Phadke

³ M. Lambla, M. Seadan, *Makromol. Chem., Macromol. Symp.* **69** (1993) 99

⁴ N. C. Liu, H. Q. Xie, W. E. Baker, *Polymer* **34** (1993) 4680

⁵ N. C. Liu, W. E. Baker, *Adv. Polym. Technol.* **11** (1992) 249

⁶ N. G. Gaylord, in: "Reactive Extrusion — Principles and Practice", M. Xanthos (Ed.), Hanser, Munich 1992, Ch. 3

⁷ A. Hogt, ANTEC technical papers, Atlanta, April 1988, p. 1478

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- ⁸ V. J. Triacca, P. E. Gloor, S. Zhu, A. N. Hrymak, A. E. Hamielec, *Polym. Eng. Sci.* **33** (1993) 445
- ⁹ R. Gallucci, R. C. Goings, *J. Appl. Polym. Sci.* **27** (1982) 425
- ¹⁰ Guo-Hua Hu, Jean-Jacques Flat, Morand Lambla, ANTEC technical papers, San Francisco, May 1994, p. 2775
- ¹¹ Y. Iwakura, K. Matsuzaki, *Kobunshi Kagaku* **17** (1960) 187
- ¹² J. A. Simms, *J. Appl. Polymer Sci.* **5** (1961) 58